EXPERIMENT 3: INDIVIDUAL VARIATIONS IN THE RESPONSE TO DRUGS

300 Level Pharmacology Practical & In-Course Examination Master Study Guide

Department of Pharmacology, College of Medicine / University Teaching Hospital
Source Document: Practical Lab Manual & Report — Page 31 (Printed Experimental Text)
Target: 30 Verbatim High-Yield In-Course Examination Questions & Model Answers

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📄 Exact Verbatim Transcription of Page 31 (Printed Text)

For complete reference and verbatim verification, here is the exact text printed on Page 31 of the laboratory manual:

EXPERIMENT 3

INDIVIDUAL VARIATIONS IN THE RESPONSE TO DRUGS

Introduction
Just as individuals vary in weight, height and other characteristics, so also do they vary in their response to drugs. A few may show totally unexpected and abnormal responses to a particular drug (idiosyncrasy). Even when these abnormalities are ignored, there is a wide individual variation in the dose of a drug needed to produce the same pharmacological or therapeutic response. For example, when sodium amobarbital (sodium amytal) was widely used for obstetrical anaesthesia and analgesia, it was discovered that even if this drug were given intravenously and on the basis of mg per kg of body weight, the patients still responded to the same dose to a different degree (Fig. 2). In the majority, to be sure, a dose of about 10mg per kg. produced satisfactory obstetrical narcosis, but in some individual this dose produced only small effects, while in others a dangerous degree of central depression. On the whole, this variation is not evenly distributed. The curve of distribution is usually not symmetrical and bell-shaped, but somewhat skewed. The right-hand of the curve shows that relatively resistant individuals and the left-hand end of the curve, the hypersusceptible ones. It can be demonstrated that when the doses are plotted logarithmically, the curve can usually be converted to a normally bell shaped frequency curve. This method is also utilized in the determination of the median lethal dose of drugs in animals.

Experimental
Drugs: 10mg/ml sodium pentobarbitone
Equipment: Beakers (250 cc); Syringes (tuberculin); needle (gauge 25).
Method: Inject each with 25mg/kg of sodium hexobarbitone intraperitoneally. Make observation as to development of ataxia, muscular relaxation, response to painful stimuli and respiration. Wait for some recovery time if necessary and then administer the next higher dose (50mg/kg hexobarbitone) to the remaining animals in the test population. Note the percentage that respond and separate them also. Repeat steps 3 and 4 with 75, 100 and 125mg/kg respectfully until the animals in the test population have responded. Plot a graph of percentage frequency of response against dose (mg/kg) and discuss the graph.

🎯 High-Yield Conceptual Summary for In-Course Exams

1. **Biological Variation vs Idiosyncrasy:**
   - **Idiosyncrasy:** A qualitatively abnormal, unexpected drug response (usually genetically determined, e.g., G6PD deficiency causing hemolysis with primaquine).
   - **Biological Variation:** A quantitative difference in the dose needed to achieve a standard response across a normal population.
2. **The Obstetrical Amobarbital Model:**
   - 10 mg/kg IV produced three distinct clinical outcomes in different patients:
     - *Majority:* Satisfactory obstetrical narcosis.
     - *Hypersusceptible (Left Tail):* Dangerous central nervous system (CNS) depression / respiratory failure.
     - *Resistant (Right Tail):* Only minimal or negligible effects.
3. **Curve Symmetry & Log Transformation:**
   - When plotted on an **arithmetic (linear) dose scale**, frequency distribution curves of drug responses are **skewed** (asymmetrical, with a prolonged right tail).
   - When plotted on a **logarithmic dose scale (log dose)**, the curve is mathematically converted into a **symmetrical, bell-shaped Gaussian normal distribution curve**.
   - This principle underpins the determination of the **ED_{50} (median effective dose)** and **LD_{50} (median lethal dose)** in bioassays.
4. **Laboratory Methodology (Quantal Barbiturate Response):**
   - **Animals Tested:** Injected intraperitoneally (IP).
   - **Drugs Listed:** 10 mg/ml sodium pentobarbitone (listed under Drugs); sodium hexobarbitone (used in method protocol).
   - **Needle/Syringe:** 25-gauge needle; tuberculin syringe (for accurate small-volume measurement).
   - **Dose Escalation:** 25 mg/kg → 50 mg/kg → 75 mg/kg → 100 mg/kg → 125 mg/kg.
   - **Four Clinical Endpoints:** (1) Ataxia, (2) Muscular relaxation, (3) Response to painful stimuli, (4) Respiration.

📝 The 30 Verbatim In-Course Examination Questions & Answers

Question 1: Experiment Title

What is the full official title of Experiment 3 as printed in the practical manual?

- **Verbatim Answer:**
  `INDIVIDUAL VARIATIONS IN THE RESPONSE TO DRUGS`
- **In-Course Context & Relevance:**
  In pharmacology practical vivas and in-course spot tests, students are frequently asked to state the

precise aim or title of the practical. This experiment illustrates quantal dose-response relationships and biological variability in animal populations.

Question 2: Biological Analogy of Drug Variation

What fundamental anatomical and physical characteristics does the text compare to variation in drug response?

- **Verbatim Answer:**
  `Just as individuals vary in weight, height and other characteristics, so also do they vary in their response to drugs.`
- **In-Course Context & Relevance:**
  Biological variation is a universal feature of living systems. Just as height and weight follow continuous distribution patterns, the threshold concentration of a drug required to elicit a defined response varies among members of a species due to genetic, pharmacokinetic, and pharmacodynamic factors.

Question 3: Definition of Idiosyncrasy

How does the introduction on Page 31 define "idiosyncrasy"?

- **Verbatim Answer:**
  `A few may show totally unexpected and abnormal responses to a particular drug (idiosyncrasy).`
- **In-Course Context & Relevance:**
  High-yield exam trap: Lecturers frequently test the distinction between tolerance, allergy, and idiosyncrasy. Idiosyncrasy is defined strictly as a genetically determined, qualitatively abnormal, uncharacteristic reaction to a drug that does not involve an allergic (immune-mediated) mechanism and is not reproducible in most individuals at any dose.

Question 4: Scope of Individual Variation

What does the text state occurs even when idiosyncratic abnormalities are completely ignored?

- **Verbatim Answer:**
  `Even when these abnormalities are ignored, there is a wide individual variation in the dose of a drug needed to produce the same pharmacological or therapeutic response.`
- **In-Course Context & Relevance:**
  This highlights that biological variability is not limited to rare pathological outliers. Even among a completely healthy, non-idiosyncratic population, the dose of a drug required to achieve an identical therapeutic endpoint varies substantially.

Question 5: Historical Clinical Drug Example

Which specific drug is cited as the historical clinical example demonstrating individual response variation?

- **Verbatim Answer:**
  `sodium amobarbital`
- **In-Course Context & Relevance:**
  Sodium amobarbital is an intermediate-acting barbiturate. Lecturers expect students to cite this exact agent when discussing the historical clinical demonstration of variable narcosis.

Question 6: Alternative Commercial/Generic Name

By what synonym or brand name in parentheses does the text identify sodium amobarbital?

- **Verbatim Answer:**
  `(sodium amytal)`
- **In-Course Context & Relevance:**
  Amytal was the commercial formulation synthesized by Eli Lilly and widely utilized in 20th-century medicine. In viva examinations, examiners may use "Sodium Amytal" and "Sodium Amobarbital" interchangeably.

Question 7: Historical Clinical Indication

For what dual clinical indications was sodium amobarbital widely used according to the text?

- **Verbatim Answer:**
  `obstetrical anaesthesia and analgesia`
- **In-Course Context & Relevance:**
  During the early-to-mid 1900s, barbiturates were administered during labor to produce "twilight sleep" or obstetrical analgesia/sedation, an application that clearly revealed variable maternal depression and fetal narcosis.

Question 8: Clinical Route of Administration

By which route of administration was sodium amobarbital administered in the obstetrical clinical example cited?

- **Verbatim Answer:**
  `intravenously`
- **In-Course Context & Relevance:**
  Intravenous administration bypasses gastrointestinal absorption variables (first-pass metabolism, gastric emptying). Therefore, the observed variations were pharmacodynamic and distribution-dependent, rather than mere absorption defects.

Question 9: Dose Calculation Basis

On what specific physiological parameter was the dose of sodium amobarbital calculated for each patient?

- **Verbatim Answer:**
  `on the basis of mg per kg of body weight`
- **In-Course Context & Relevance:**
  Even after standardizing dosage by body weight (mg/kg)—which eliminates simple weight differences as the source of error—wide variation in patient response persisted.

Question 10: Clinical Finding of Standardized Dosing

What was discovered when sodium amobarbital was administered on a standardized mg/kg basis?

- **Verbatim Answer:**
  `it was discovered that even if this drug were given intravenously and on the basis of mg per kg of body weight, the patients still responded to the same dose to a different degree (Fig. 2).`
- **In-Course Context & Relevance:**
  This proved conclusively that body weight alone cannot predict drug sensitivity; receptor density, central nervous system excitability, plasma protein binding, and metabolic enzyme activity create distinct response thresholds.

Question 11: Standard Therapeutic Dose

In the majority of patients, what specific dose of sodium amobarbital produced "satisfactory obstetrical narcosis"?

- **Verbatim Answer:**
  `a dose of about 10mg per kg.`
- **In-Course Context & Relevance:**
  10 mg/kg was established as the median effective dose (ED_{50}) for obstetrical narcosis in that clinical cohort.

Question 12: Response of Hypo-Responders at 10 mg/kg

What degree of response did this 10 mg/kg dose produce in the resistant minority of individuals?

- **Verbatim Answer:**
  `in some individual this dose produced only small effects`
- **In-Course Context & Relevance:**
  These resistant individuals represent the right-hand tail of the distribution curve; their central nervous system receptors or rapid metabolic clearance require significantly higher doses to achieve clinical sedation.

Question 13: Response of Hypersusceptible Individuals at 10 mg/kg

What dangerous outcome did the standard 10 mg/kg dose produce in sensitive individuals at the other extreme?

- **Verbatim Answer:**
  `while in others a dangerous degree of central depression.`
- **In-Course Context & Relevance:**
  This life-threatening central depression (respiratory arrest, profound coma) in hypersusceptible patients illustrates the narrow safety margin of barbiturates and why fixed-dose therapeutic protocols can be lethal without individualized titration.

Question 14: Distribution of Variation

How does the text describe the distribution of individual variation across a population?

- **Verbatim Answer:**
  `On the whole, this variation is not evenly distributed.`
- **In-Course Context & Relevance:**
  Drug sensitivity does not follow a simple uniform or evenly spaced step-wise progression; instead, clustering occurs around the median, with tapering asymmetrical frequencies at the extremes.

Question 15: Arithmetic Curve Shape

What two geometric characteristics does the distribution curve usually NOT exhibit when plotted directly against linear arithmetic doses?

- **Verbatim Answer:**
  `The curve of distribution is usually not symmetrical and bell-shaped`
- **In-Course Context & Relevance:**
  Unlike idealized Gaussian curves, raw biological drug sensitivity curves on an arithmetic axis are typically skewed because response thresholds cluster at lower doses while resistant outliers extend far to the right.

Question 16: Actual Shape of the Raw Dose Curve

How does the text specifically describe the actual shape of the distribution curve?

- **Verbatim Answer:**
  `somewhat skewed`
- **In-Course Context & Relevance:**
  The curve exhibits positive (rightward) skewness. In mathematical pharmacology, biological systems naturally display log-normal, rather than normal, distribution of drug sensitivity.

Question 17: Right-Hand End of the Distribution Curve

What group of individuals is situated at the right-hand end of the distribution curve?

- **Verbatim Answer:**
  `The right-hand of the curve shows that relatively resistant individuals`
- **In-Course Context & Relevance:**
  The right tail contains resistant subjects who tolerate exceptionally high doses before exhibiting the

quantal endpoint. In clinical practice, treating these patients requires dose escalation.

Question 18: Left-Hand End of the Distribution Curve

What group of individuals is situated at the left-hand end of the distribution curve?

- **Verbatim Answer:**
  `and the left-hand end of the curve, the hypersusceptible ones.`
- **In-Course Context & Relevance:**
  The left tail contains hypersusceptible (hyper-reactive) individuals who develop profound pharmacological and adverse effects at doses far below the standard median therapeutic dose.

Question 19: Logarithmic Transformation of the Dose Curve

How can the skewed distribution curve be converted into a normally bell-shaped frequency curve?

- **Verbatim Answer:**
  `It can be demonstrated that when the doses are plotted logarithmically, the curve can usually be converted to a normally bell shaped frequency curve.`
- **In-Course Context & Relevance:**
  Plotting response against the logarithm of dose (\log \text{dose}) compresses the elongated right tail and expands the left tail, transforming a skewed biological curve into a symmetrical Gaussian (bell-shaped) normal distribution curve. This is the cornerstone of modern pharmacodynamics and probit analysis.

Question 20: Preclinical Application of the Log Method

In what major pharmacological determination in animals is this logarithmic curve conversion method utilized?

- **Verbatim Answer:**
  `This method is also utilized in the determination of the median lethal dose of drugs in animals.`
- **In-Course Context & Relevance:**
  In preclinical toxicology, determining the LD_{50} (median lethal dose) requires plotting quantal mortality data logarithmically to establish symmetrical probit lines, allowing statistical calculation of drug safety and Therapeutic Index (TI = LD_{50} / ED_{50}).

Question 21: Full Formal Name of LD_{50}

What is the full formal name of the pharmacological parameter abbreviated as LD_{50} in animal studies?

- **Verbatim Answer:**
  `median lethal dose`
- **In-Course Context & Relevance:**
  The median lethal dose (LD_{50}) is the statistically calculated dose of a chemical or drug that will kill exactly 50% of an animal test population under specified experimental conditions.

Question 22: Experimental Drug Listed under "Drugs"

What specific drug formulation and concentration are listed under the "Drugs" heading in the Experimental section?

- **Verbatim Answer:**
  `10mg/ml sodium pentobarbitone`
- **In-Course Context & Relevance:**
  High-yield exam spotter: The manual explicitly specifies "10mg/ml sodium pentobarbitone" in the materials list. Pentobarbitone (pentobarbital) is a short-acting oxybarbiturate commonly used in veterinary and laboratory rodent anesthesia.

Question 23: Beaker Volume Specification

What specific volume and type of glassware are designated under the "Equipment" section?

- **Verbatim Answer:**
  `Beakers (250 cc)`
- **In-Course Context & Relevance:**
  250 cc (cubic centimeters, equivalent to 250 mL) glass beakers are used to hold, invert, or observe individual small rodents during behavioral sedation testing.

Question 24: Syringe Type Specification

What specific type of syringe is mandated in the Equipment list?

- **Verbatim Answer:**
  `Syringes (tuberculin)`
- **In-Course Context & Relevance:**
  Tuberculin syringes are calibrated 1.0 mL syringes with fine gradations (0.01 mL). They are indispensable in rodent pharmacology to accurately measure minute injection volumes (e.g., 0.05 to 0.1 mL) based on animal body weight.

Question 25: Needle Gauge Specification

What exact needle gauge is required for drug injection in this experiment?

- **Verbatim Answer:**
  `needle (gauge 25)`
- **In-Course Context & Relevance:**
  A 25-gauge (25G) needle has a narrow outer diameter (~0.5 mm) suitable for atraumatic intraperitoneal (IP) puncture in small laboratory rodents (mice and rats) without lacerating abdominal viscera.

Question 26: Initial Drug, Starting Dose, and Route

According to Step 1 of the Method, what drug, initial dose, and route of administration are specified?

- **Verbatim Answer:**
  `Inject each with 25mg/kg of sodium hexobarbitone intraperitoneally.`
- **In-Course Context & Relevance:**
  Exam focus: (1) Drug: sodium hexobarbitone (a very short-acting barbiturate), (2) Dose: 25 mg/kg, (3) Route: intraperitoneally (IP). Intraperitoneal injection allows rapid absorption through the extensive vascular surface of the peritoneal mesentery into portal circulation.

Question 27: Four Mandatory Clinical Observations

Which four specific physiological and neurological parameters must be observed following injection?

- **Verbatim Answer:**
  `Make observation as to development of ataxia, muscular relaxation, response to painful stimuli and respiration.`
- **In-Course Context & Relevance:**
  These four signs track progressive stages of central nervous system (CNS) depression:
  1. *Ataxia:* Impaired motor coordination (cerebellar/vestibular depression).
  2. *Muscular relaxation:* Loss of postural muscle tone (spinal cord/reticular suppression).
  3. *Response to painful stimuli:* Analgesia/loss of pedal reflex (thalamocortical depression).
  4. *Respiration:* Respiratory rate and depth (medullary depression).

Question 28: Procedure Between Dose Increments

What action must be taken before administering the next higher dose to non-responding animals?

- **Verbatim Answer:**
  `Wait for some recovery time if necessary and then administer the next higher dose (50mg/kg hexobarbitone) to the remaining animals in the test population.`
- **In-Course Context & Relevance:**
  Allowing recovery or ensuring that subthreshold effects have stabilized prevents erratic cumulative toxicity and ensures that only truly resistant non-responders receive the escalating dose.

Question 29: Subsequent Escalating Dose Sequence

What are the exact subsequent escalating doses (in mg/kg) administered to non-responding animals?

- **Verbatim Answer:**
  `Repeat steps 3 and 4 with 75, 100 and 125mg/kg respectfully until the animals in the test population have responded.`
- **In-Course Context & Relevance:**
  The full arithmetic dose series in the protocol is: **25 mg/kg → 50 mg/kg → 75 mg/kg → 100 mg/kg → 125 mg/kg** (increasing by increments of 25 mg/kg). Note the manual's typographical use of "respectfully" for "respectively"—an exact verbatim detail!

Question 30: Final Graph Plotting Requirement

What specific graph must be plotted and analyzed at the conclusion of the experiment?

- **Verbatim Answer:**
  `Plot a graph of percentage frequency of response against dose (mg/kg) and discuss the graph.`
- **In-Course Context & Relevance:**
  Plotting Percentage Frequency of Response (Y-axis) versus Dose in mg/kg (X-axis) generates an empirical quantal frequency distribution curve. Discussing the graph requires identifying the median effective dose (ED_{50}), evaluating curve skewness, and explaining how biological variation accounts for non-uniform response across animal cohorts.

🏆 In-Course Quick-Review Master Cheat Sheet

Feature / Question	Verbatim Manual Fact (Page 31)	In-Course Clinical Pearl
Experiment Title	Individual Variations in the Response to Drugs	Quantal dose-response bioassay
Abnormal Response	Idiosyncrasy	Genetically determined, qualitative abnormality
Historical Drug	Sodium amobarbital (sodium amytal)	Intermediate-acting barbiturate (GABA_A positive modulator)
Historical Use	Obstetrical anaesthesia and analgesia	Sedation and analgesia in active labor
Clinical Route	Intravenously (IV)	Eliminates GI absorption variation
Dose Calculation	mg per kg of body weight	Standardized for body mass
Standard Dose	10 mg/kg	Produced satisfactory obstetrical narcosis in the majority
Resistant Response	Produced only small effects	Right tail of distribution curve
Sensitive Response	Dangerous degree of central depression	Left tail of distribution curve (respiratory arrest risk)
Linear Curve Shape	Not symmetrical / somewhat skewed	Positive (rightward) skewness on arithmetic axis
Symmetrical Curve	Plotted logarithmically	Becomes normally bell-shaped Gaussian curve
Toxicology Use	Determination of median lethal dose (LD_{50})	Probit analysis in animal safety assays
Drug in "Drugs"	10mg/ml sodium pentobarbitone	Short-acting barbiturate formulation
Equipment	Beakers (250 cc), Syringes (tuberculin), Needle (gauge 25)	Tuberculin for small volumes, 25G for atraumatic IP puncture
Method Drug	Sodium hexobarbitone	Ultra-short acting anesthetic barbiturate
Initial Dose & Route	25 mg/kg intraperitoneally (IP)	Rapid mesenteric capillary absorption
4 Observations	Ataxia, muscular relaxation, pain response, respiration	Progressive signs of CNS & medullary depression
Dose Escalation	25 → 50 → 75 → 100 → 125 mg/kg	25 mg/kg incremental quantal escalation
Graph Requirement	Percentage frequency of response vs dose (mg/kg)	Quantal dose-response frequency curve